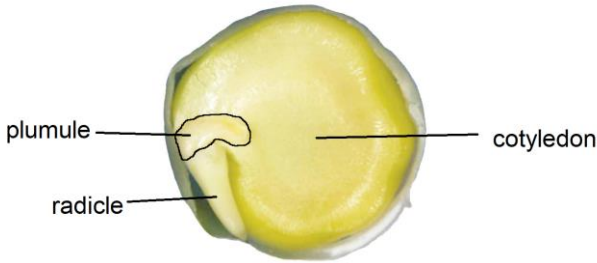


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		 all 3 labels correct = 2 marks 2 labels correct = 1 mark		2		2		2
		(ii)		• {Anchors / roots / stabilises} in {soil / earth / ground} (1) • for {water / minerals / nutrients} (1)		2		2		
	(b)			Any three (x1) from A. water is {taken up / imbibed / owtte} through the micropyle (1) B. the testa {splits / owtte} (1) C. (which allows) oxygen (to enter / used in respiration) (1) D. {food reserves / starch / proteins} (from the cotyledons) are {hydrolysed / digested / broken down / owtte} (1)	3			3		
	(c)			Any three (x1) from ABA not produced so: A. gibberellin diffuses to the aleurone layer (1) B. gibberellins {induce / cause the production / cause the release} of {hydrolytic enzymes / amylase / named enzyme} (1) C. {glucose / other nutrients} are used in {(aerobic) respiration / growth} (1) D. {Roots / radicle / coleoptile / plumule} {have grown / are visible} (1)		3		3		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		<i>The idea that any of the following may not be the same in each field</i> • <u>soil</u> pH • water supply / rainfall • sunlight / exposure to light / light intensity / shading • (soil) nutrients / minerals / named nutrients / fertility / owtte • Waterlogging / aeration of soil / oxygen content <u>of soil</u> / • Presence of weeds / pests / slugs / snails / rabbits / disease • Clay vs sandy soil / compact vs tilled / owtte Ignore humidity or type of soil (unless qualified) four correct = 2 marks 2/3 correct = 1 mark 0/1 correct = 0 marks		2		2		2
		(ii)		• Any reference to global warming / climate change / owtte (1) • Reduced {crop / yield / wheat} (1) ignore less wheat plants grown • Less for {food / feeding animals} / leads to famine / leads to food insecurity (1)			3	3		
				Question 2 total	3	9	3	15	0	4